

Data Science — All Units MCQ Revision

Questions + Correct Answers Only

UNIT 1 — Introduction to Data Science

Q1. What is the primary goal of data science?

Correct Answer: To extract insights from data

Q2. Which of the following is NOT a step in the data science process?

Correct Answer: Data Ignoring

Q3. Which of the following techniques is commonly used in machine learning?

Correct Answer: Linear Regression

Q4. What does EDA stand for?

Correct Answer: Exploratory Data Analysis

Q5. Which programming language is most commonly associated with data science?

Correct Answer: Python

Q6. What is the purpose of data visualization?

Correct Answer: To present data in a clear and understandable way

Q7. Which of the following is a type of supervised learning?

Correct Answer: Classification

Q8. What is a primary benefit of data-driven decision making?

Correct Answer: Making decisions based on empirical evidence

Q9. How does data science help in understanding customer behaviour?

Correct Answer: By analysing customer data for insights

Q10. What is predictive analytics primarily used for?

Correct Answer: To forecast future trends and outcomes

Q11. Which of the following is a way data science can improve operational efficiency?

Correct Answer: By identifying inefficiencies and suggesting improvements

Q12. What does gaining a competitive advantage through data science involve?

Correct Answer: Using insights that competitors do not have

Q13. What role does data science play in innovation?

Correct Answer: It enables exploration of new business models based on data insights

Q14. Which of the following industries has benefited from data science?

Correct Answer: All of the above

Q15. What is the primary goal of Business Intelligence?

Correct Answer: Monitoring and reporting historical data

Q16. Which tool is commonly used in Data Science?

Correct Answer: Python

Q17. Business Intelligence focuses primarily on:

Correct Answer: Data visualization and dashboards

Q18. What is the key difference between BI and Data Science?

Correct Answer: BI provides descriptive analytics, while Data Science offers predictive insights

Q19. Which of the following is a predictive analytics technique?

Correct Answer: Time-series forecasting

Q20. What is the first step in the Data Science process?

Correct Answer: Data Collection

Q21. Which of the following is NOT a part of Data Preparation?

Correct Answer: Model deployment

Q22. What is the primary goal of Exploratory Data Analysis (EDA)?

Correct Answer: Identify patterns and trends in data

Q23. Which of these tools is widely used for Data Modelling in Data Science?

Correct Answer: Scikit-learn

Q24. Which metric is used to evaluate the accuracy of a classification model?

Correct Answer: Precision

Q25. What is the main objective of Data Visualization?

Correct Answer: Communicate insights effectively

Q26. Which is the first step in the Data Science Life Cycle?

Correct Answer: Problem Definition

Q27. What is the main objective of Data Preparation?

Correct Answer: Cleaning and organizing data for analysis

Q28. Which process helps identify patterns and trends in data?

Correct Answer: Exploratory Data Analysis (EDA)

Q29. What step involves splitting data into training and testing datasets?

Correct Answer: Data Preparation

Q30. Which of the following is NOT a model evaluation metric?

Correct Answer: Data Cleaning

Q31. After deploying a model, what is the next step?

Correct Answer: Monitoring and Maintenance

Q32. What is the purpose of data visualization during EDA?

Correct Answer: Identify patterns and insights

Q33. Which of the following is NOT a data storage tool?

Correct Answer: Matplotlib

Q34. Which language is widely used for data manipulation and analysis?

Correct Answer: Python

Q35. Tableau is primarily used for:

Correct Answer: Data Visualization

Q36. Which framework is used for building machine learning models?

Correct Answer: TensorFlow

Q37. Which of the following is a version control tool?

Correct Answer: Git

Q38. For processing large-scale data in a distributed environment, which tool is best suited?

Correct Answer: Apache Spark

Q39. What tool is commonly used for creating dashboards and business intelligence reports?

Correct Answer: Power BI

Q40. Which Python library is used for data visualization?

Correct Answer: Matplotlib

Q41. Docker is used for:

Correct Answer: Model Deployment

Q42. Jupyter Notebook is mainly used for:

Correct Answer: Writing code and documenting analysis

UNIT 2 — Big Data

Q43. What type of Big Data is an email?

Correct Answer: Unstructured

Q44. Which of the following is an example of structured data?

Correct Answer: Relational database records

Q45. What type of Big Data is generated by IoT devices?

Correct Answer: Machine-generated

Q46. Which classification applies to Big Data processed in real-time?

Correct Answer: Stream Data

Q47. XML and JSON are examples of which type of Big Data?

Correct Answer: Semi-structured

Q48. What is Big Data primarily defined by?

Correct Answer: High volume, velocity, and variety

Q49. When did Big Data gain prominence as a field of study?

Correct Answer: 2000s

Q50. Which of the following technologies played a key role in the evolution of Big Data?

Correct Answer: Both A and C — Apache Hadoop and MapReduce

Q51. What triggered the need for new Big Data tools in the 2000s?

Correct Answer: Increase in social media and IoT-generated data

Q52. Which of the following best describes the modern era of Big Data?

Correct Answer: Integration of Big Data with AI, IoT, and cloud computing

Q53. What is the main function of the Data Ingestion Layer in Big Data architecture?

Correct Answer: Importing and capturing data from sources

Q54. Which of the following technologies is used for real-time data processing in Big Data architecture?

Correct Answer: Apache Spark

Q55. What type of storage is HDFS primarily designed for?

Correct Answer: Distributed storage

Q56. Which layer in Big Data architecture is responsible for analyzing and visualizing the data?

Correct Answer: Data Analysis and Query Layer

Q57. What is the main purpose of NoSQL databases in Big Data architecture?

Correct Answer: Handling unstructured and semi-structured data

UNIT 3 — HADOOP

Q58. What is the main purpose of Hadoop?

Correct Answer: To process and store large datasets in a distributed manner

Q59. What is HDFS in Hadoop?

Correct Answer: Hadoop Distributed File System

Q60. Which of the following is NOT a component of Hadoop?

Correct Answer: SQL Server

Q61. What does YARN stand for?

Correct Answer: Yet Another Resource Negotiator

Q62. What is the role of MapReduce in Hadoop?

Correct Answer: To process data in parallel

Q63. How does Hadoop handle hardware failures?

Correct Answer: By replicating data across multiple nodes

Q64. Which programming language is primarily used for Hadoop?

Correct Answer: Java

Q65. What is the default replication factor in HDFS?

Correct Answer: 3

Q66. Which tool in the Hadoop ecosystem is used for SQL-like queries?

Correct Answer: Hive

Q67. What is the role of Zookeeper in the Hadoop ecosystem?

Correct Answer: Managing and coordinating distributed applications

Q68. What is the primary function of HDFS in Hadoop?

Correct Answer: To store data in a distributed manner

Q69. What is the responsibility of the NameNode in HDFS?

Correct Answer: Manage metadata and block locations

Q70. Which component is responsible for resource allocation in YARN?

Correct Answer: ResourceManager

Q71. What are the two phases of a MapReduce job?

Correct Answer: Map and Reduce

Q72. Which tool in the Hadoop ecosystem is used for real-time data processing?

Correct Answer: Spark

Q73. What is the default block size in HDFS for storing data?

Correct Answer: 128MB

Q74. Which Hadoop ecosystem component is used for machine learning?

Correct Answer: Mahout

Q75. Which of the following tools is used for importing and exporting data in Hadoop?

Correct Answer: Sqoop

Q76. In the Hadoop architecture, which node performs data storage tasks?

Correct Answer: DataNode

Q77. What is the role of Zookeeper in the Hadoop ecosystem?

Correct Answer: Coordination and synchronization

Q78. Which Hadoop tool provides SQL-like querying capabilities?

Correct Answer: Hive

Q79. What is the purpose of HBase in the Hadoop ecosystem?

Correct Answer: To provide NoSQL database storage

Q80. Which component is used for data import/export between Hadoop and relational databases?

Correct Answer: Sqoop

Q81. What is the main use of Flume in the Hadoop ecosystem?

Correct Answer: To collect and move log data

Q82. Which of the following is a workflow management tool in Hadoop?

Correct Answer: Oozie

Q83. What is the primary function of Spark in the Hadoop ecosystem?

Correct Answer: In-memory data processing

Q84. Which tool is used for machine learning tasks in Hadoop?

Correct Answer: Mahout

Q85. What does Zookeeper provide in the Hadoop ecosystem?

Correct Answer: Data synchronization and coordination

Q86. Which component handles the streaming of real-time data in Hadoop?

Correct Answer: Kafka

Q87. Which tool in the Hadoop ecosystem is ideal for large-scale graph processing?

Correct Answer: Spark

Q88. What is the main purpose of MapReduce?

Correct Answer: To process large datasets in a distributed manner

Q89. What does the Mapper in MapReduce do?

Correct Answer: Processes input data to produce key-value pairs

Q90. What is the role of the Reducer in MapReduce?

Correct Answer: To aggregate and summarize key-value pairs

Q91. What phase in MapReduce organizes key-value pairs by key?

Correct Answer: Shuffle and Sort phase

Q92. Which of the following is NOT a part of the MapReduce framework?

Correct Answer: NameNode

Q93. What is the Combiner used for in MapReduce?

Correct Answer: To reduce the amount of data transferred to the Reducer

Q94. Which of the following describes the output of the Map phase?

Correct Answer: Intermediate key-value pairs

Q95. In MapReduce, what is the default number of Reducers?

Correct Answer: 1

Q96. What type of tasks can MapReduce efficiently handle?

Correct Answer: Batch processing of large-scale datasets

Q97. Which Hadoop component is responsible for scheduling MapReduce tasks?

Correct Answer: YARN

Q98. What is the primary function of HDFS?

Correct Answer: To store large datasets in a distributed manner

Q99. Which component of HDFS manages metadata?

Correct Answer: NameNode

Q100. What is the default block size in HDFS?

Correct Answer: 128MB

Q101. What does the Secondary NameNode do?

Correct Answer: Periodically merges and checkpoints metadata

Q102. How does HDFS ensure fault tolerance?

Correct Answer: By replicating data blocks across multiple nodes

Q103. What is the default replication factor in HDFS?

Correct Answer: 3

Q104. Which of the following is NOT a characteristic of HDFS?

Correct Answer: Real-time data updates

Q105. What happens if a DataNode fails in HDFS?

Correct Answer: Data is retrieved from replicated blocks on other nodes

Q106. Which node in HDFS stores actual data?

Correct Answer: DataNode

Q107. What is the primary function of YARN in Hadoop?

Correct Answer: Resource management and task scheduling

Q108. Which component of YARN manages resources on individual nodes?

Correct Answer: NodeManager

Q109. What is a container in YARN?

Correct Answer: A unit of resource allocation for tasks

Q110. What is the role of the ResourceManager in YARN?

Correct Answer: To manage resources across the cluster

Q111. Which YARN component manages the lifecycle of an application?

Correct Answer: ApplicationMaster

Q112. What happens if the NodeManager fails?

Correct Answer: Tasks on the node are rescheduled on other nodes

Q113. Which of the following is NOT a feature of YARN?

Correct Answer: Data replication

Q114. What does YARN allocate to applications for execution?

Correct Answer: Containers

Q115. How does YARN improve resource utilization?

Correct Answer: By dynamically allocating resources based on demand

Q116. Which processing frameworks can run on YARN?

Correct Answer: Any framework designed for distributed processing

UNIT 4 — Advance Database System

Q117. Which database is based on a table structure of rows and columns?

Correct Answer: Relational Database

Q118. MongoDB is an example of which type of database?

Correct Answer: NoSQL Database

Q119. Which type of database is best suited for handling relationships in social networks?

Correct Answer: Graph Database

Q120. What is the primary feature of a distributed database?

Correct Answer: Data distributed across multiple systems

Q121. AWS RDS is an example of a:

Correct Answer: Cloud Database

Q122. Which of the following best describes NoSQL databases?

Correct Answer: They provide support for unstructured or semi-structured data

Q123. Which type of NoSQL database is optimized for managing relationships between data?

Correct Answer: Graph Database

Q124. MongoDB is an example of which type of NoSQL database?

Correct Answer: Document Store

Q125. Which of the following is a feature of NoSQL databases?

Correct Answer: Support for distributed data

Q126. Cassandra is an example of which type of NoSQL database?

Correct Answer: Column-Family Store

Q127. Why are NoSQL databases preferred for handling big data?

Correct Answer: They can process large volumes of unstructured data

Q128. What makes NoSQL databases more flexible compared to relational databases?

Correct Answer: Schema-less data modeling

Q129. What type of scalability is a key feature of NoSQL databases?

Correct Answer: Horizontal scalability

Q130. Which of the following is NOT a reason for using NoSQL databases?

Correct Answer: Fixed schema requirement

Q131. For real-time applications requiring high-speed data access, which type of database is suitable?

Correct Answer: NoSQL databases

Q132. What is the main advantage of NoSQL databases in terms of scalability?

Correct Answer: Horizontal scaling

Q133. Which of the following data types can NoSQL databases handle?

Correct Answer: Structured, semi-structured, and unstructured data

Q134. Why are NoSQL databases suitable for real-time applications?

Correct Answer: They are optimized for high-speed read and write operations

Q135. Which feature of NoSQL databases reduces the complexity of data retrieval?

Correct Answer: No need for complex joins

Q136. What makes NoSQL databases cost-effective?

Correct Answer: Many are open-source and use commodity hardware

Q137. How do NoSQL databases handle schema changes?

Correct Answer: Automatically adapts to schema changes

Q138. Which type of database uses a predefined schema?

Correct Answer: SQL

Q139. What is a major advantage of NoSQL databases over SQL databases?

Correct Answer: Horizontal scalability

Q140. Which type of database is better suited for handling structured data?

Correct Answer: SQL

Q141. Which of the following is an example of a NoSQL database?

Correct Answer: MongoDB

Q142. What makes NoSQL databases more flexible than SQL databases?

Correct Answer: Schema-less design

Q143. SQL databases are typically scaled by:

Correct Answer: Adding more resources to the same server (vertical scaling)

Q144. Which type of database is ideal for applications with rapidly changing data models?

Correct Answer: NoSQL

Q145. Which query language is used by SQL databases?

Correct Answer: Structured Query Language (SQL)

Q146. Which type of NoSQL database stores data in key-value pairs?

Correct Answer: Key-Value Store

Q147. MongoDB is an example of which type of NoSQL database?

Correct Answer: Document Store

Q148. Which type of NoSQL database is optimized for managing relationships between entities?

Correct Answer: Graph Database

Q149. What type of NoSQL database is best suited for analytics and time-series data?

Correct Answer: Column-Family Store

Q150. Redis is an example of which type of NoSQL database?

Correct Answer: Key-Value Store

Q151. Which type of NoSQL database is most suitable for storing hierarchical data in JSON format?

Correct Answer: Document Store

Q152. Which of the following NoSQL databases is a graph database?

Correct Answer: Neo4j

Q153. Which NoSQL database type is best for storing large-scale tabular data?

Correct Answer: Column-Family Store

UNIT 5 — Data Analytics

Q154. What is the primary goal of data analytics?

Correct Answer: To extract valuable insights from data

Q155. Which of the following is a key step in the data analytics process?

Correct Answer: All of the above

Q156. What type of analysis is used to make predictions based on historical data?

Correct Answer: Predictive analysis

Q157. Which of the following tools is commonly used for data visualization?

Correct Answer: All of the above

Q158. Which of these is a type of unstructured data?

Correct Answer: A text document

Q159. What is the purpose of data cleaning in the analytics process?

Correct Answer: To ensure data is accurate and consistent

Q160. How can data analytics be used in sports?

Correct Answer: To improve team performance and player health

Q161. What is the first step in the Data Analytics Life Cycle?

Correct Answer: Problem Definition

Q162. In which stage of the Data Analytics Life Cycle is data cleaned and transformed for analysis?

Correct Answer: Data Cleaning and Preprocessing

Q163. Which of the following is the primary goal of Exploratory Data Analysis (EDA)?

Correct Answer: To visualize and summarize the data to find patterns and insights

Q164. At which stage of the Data Analytics Life Cycle is the model's performance evaluated?

Correct Answer: Model Evaluation

Q165. Which of the following describes the deployment and monitoring stage of the Data Analytics Life Cycle?

Correct Answer: Putting the model into use and tracking its performance

Q166. What is the purpose of the "Problem Definition" stage in the Data Analytics Life Cycle?

Correct Answer: To identify the specific questions to answer and goals to achieve

Q167. Which type of analytics answers the question, "What happened?"

Correct Answer: Descriptive Analytics

Q168. What is the primary goal of diagnostic analytics?

Correct Answer: To understand why something happened

Q169. Which type of analytics is used to predict future outcomes based on historical data?

Correct Answer: Predictive Analytics

Q170. Which of the following is a feature of prescriptive analytics?

Correct Answer: It recommends the best course of action to optimize outcomes

Q171. Which type of analytics involves using AI to simulate human thought processes?

Correct Answer: Cognitive Analytics

Q172. What is the key difference between descriptive and diagnostic analytics?

Correct Answer: Descriptive analytics summarizes historical data, while diagnostic analytics explores the reasons behind past events